Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S1 (S) Examination May 2025 (2024 Scheme)

Course Code: GXEST104

Course Name: INTRODUCTION TO ELECTRICAL AND ELECTRONICS ENGINEERING

Max. Marks: 60 Duration: 2 hour 30 minutes

- ***Use separate answer sheets for Part 1 and Part 2***
- ***No separate minimum marks are required to pass.***

PART 1:

ELECTRICAL ENGINEERING (30 Marks)

PART 1-A

Answer all questions. Each question carries 3 marks

		CO	Marks
1	State and explain Kirchhoff's laws.	1	3
2	Define the following terms i) Magnetic Flux density ii) Magneto motive force iii) Magnetic field Strength	2	3
3	Define active power, reactive power, and apparent power.	1	3
4	Show that for a sinusoidal voltage RMS value is 0.707 times its maximum value	1	3

PART 1-B

Answer any one full question from each module. Each question carries 9 marks

Module 1

5	In the circuit of Fig. 1, find the mesh currents I_1, I_2, I_3	1	9

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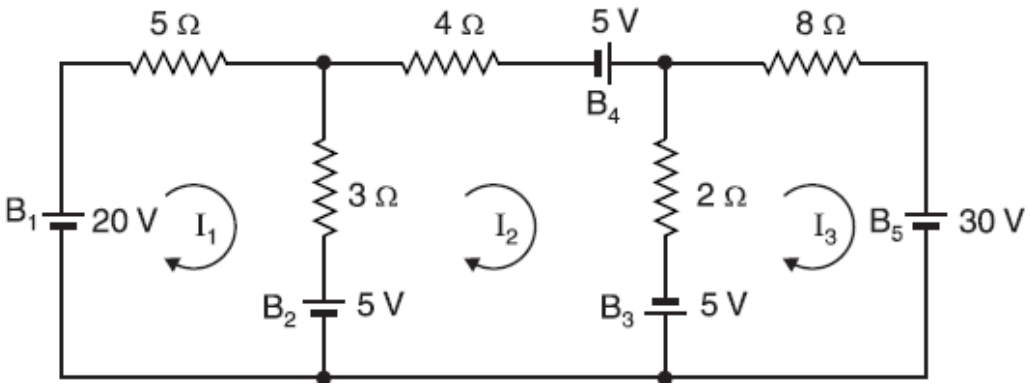


Fig.1

Calculate the equivalent resistance between the points A and B of Fig.2.

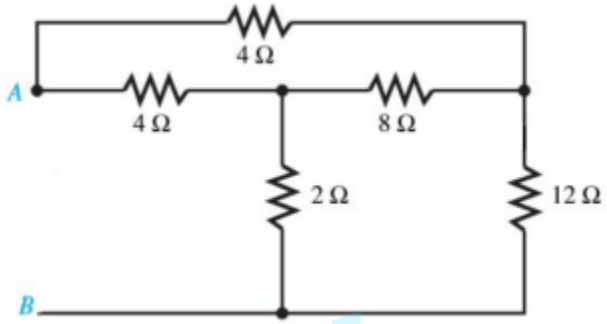


Fig.2

6	a		1	6
	b	Define reluctance. List the factors on which the reluctance of a magnetic material depends.	2	3
		Module 2		
7	a	Define self-inductance of a coil. Derive an expression for the self-inductance.	2	3
	b	A coil of resistance 12 Ω and inductive reactance of 25 Ω is connected in series with a capacitive reactance of 41 Ω. The combination is connected to a supply of 230 V, 50Hz. Find (i) circuit impedance (ii) current and (iii) power consumed.	1	6
8	a	A 3-phase, 400 V, 50 Hz a.c. supply is feeding a 3-phase delta-connected load with each phase having a resistance of 25 Ω, an inductance of 0.15 H and a capacitor of 120 μF in series. Find impedance, power factor, line current, apparent power, active power, and reactive power.	3	6
	b	List any three advantages of three-phase system over single-phase system	3	3

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PART 2

ELECTRONICS ENGINEERING (30 Marks)

PART 2-A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		Draw the block diagram of a DC power supply and explain	4	3
2		Differentiate between Zener and Avalanche breakdown	4	3
3		With the help of a neat block diagram, explain the components of a basic communication system	5	3
4		Explain the need for modulation	5	3

PART 2-B

Answer any one full question from each module. Each question carries 9 marks

		Module 3		
5	a)	Explain the construction and working of an n-channel enhancement type MOSFET. Use figures wherever necessary	4	7
	b)	Write the relation between α, the current gain CB configuration and β, the current gain in CE configuration.	4	2
6	a)	Draw the frequency response of an RC coupled amplifier. Explain why the gain is reduced at low and high frequencies	4	5
	b)	Explain the working of a PN junction diode when it is forward biased.	4	4
		Module 4		
7	a)	Explain the working of an FM receiver based on superheterodyne principle.	5	6
	b)	What are the advantages of optical communication?	5	3
8	a)	What are the essential components of a CRO?	6	3
	b)	With the help of a neat block diagram, explain the components of an electronic instrumentation system	6	6

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